Control Systems Laboratory, 1 credit

1 Course information:

Enhance student awareness of control systems by providing hands-on experience with representative aerospace systems. Design of a control system from start to finish by mapping requirements into control solutions through the process of modeling, identification, and controller design. The course has six experiments, four experiments involve an electric motor with a rotary pendulum, and the last two use a half Quad-rotor. The lab uses the Qube Servo 2 and Aero 2 equipment from Quanser Inc.

2 Course Description

There will be six laboratory experiments throughout the semester.

1. **The Qube Servo motor.** The first experiment involves studying the properties of a servo motor. The equations of motion for the servo motor are derived and the students calculate some of the parameters. This experiment shows how Coulomb friction distorts the linear solution. It also identifies the Coulomb friction in the motor. Saturation in the motor is also discussed. The students will also design and test a low-pass second-order filter to smooth out noisy data.

2. **PID and state space control of the motor.**

   The second laboratory experiment involves designing a PV controller (a particular case of a PID) and a state space controller to have the angle of the motor track a reference signal. The students are introduced to pole placement and LQR controllers. An integral state space controller is also used to track the reference signal.

3. **The rotary inverted pendulum.** This laboratory experiment introduces and uses state space feedback to have the rotary inverted pendulum follow a square wave. The students use pole placement, LQR, and a state space integral controller to stabilize the inverted rotary pendulum and track square and sinusoidal waves.

4. **The swing-up rotary pendulum.** An introduction to nonlinear control is presented. The students will use the Lyapunov function determined by the Hamiltonian, to derive a nonlinear controller to swing the pendulum from the downward position to the upward position. Then, they will also use a linear controller to hold the pendulum upright.

5. **Identification of the half quad rotor.** This laboratory experiment uses system identification techniques (with special emphasis in the Moore-Penrose pseudo inverse) to experimentally determine the parameters in the equations of motion for Quanser's Aero 2 half quad rotor system. This includes identifying the corresponding damping coefficients, Coulomb friction and motor gains for the half quad rotor system.

6. **State feedback control of the half quad rotor (helicopter configuration).**

This laboratory experiment uses pole placement, LQR, and integral controllers to control a two-degree-of-freedom helicopter and have the helicopter track a reference signal. The students apply and test their understanding of the techniques developed in previous labs by implementing a model for the helicopter and optimizing it experimentally. The end goal of this experiment is to control the helicopter and use a laser pointer attached to it to create drawings on a fluorescent screen.

Note: After all the labs are finished, the rest of the semester will be makeup labs.

3 Learning Resources and Reference Text

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: The lab manuals and their lab report templates are posted on Brightspace in a .pdf file free of charge.

4 Assignments

The students will perform 6 hands on lab experiments. Many of the lab experiments have a pre lab which must be finished before they start the lab. The students must write up their lab reports following the lab template. The lab reports will be handed in by Gradescope or Brightspace. No late lab report will be accepted unless there is a medical or university related reason.

- There are no exams or quizzes in this course.
- Lab reports are expected to be well-written, organized, and easy to read (typing is highly encouraged).
- All plots and figures in the reports should be properly labeled.
- All statements must be supported by appropriate equations, data, and both plotted and fitted results. Simply listing the output values of a fit without providing the corresponding plot is insufficient.
- Prelabs must be completed **before** the lab and uploaded to Brightspace.
- Individual lab reports are due exactly one week after the lab session. Prelabs and lab reports must be turned in via Gradescope.

5 Course Learning Objectives

On completing this course the student shall be able to:

1. Understand how Coulomb friction and other nonlinear dynamics play a role in modeling control systems and robotics.
2. Design feedback controllers to stabilize and improve the preform of certain nonlinear equipment.
3. Design control system on real world equipment.
4. Tune the controller to improve the performance of the equipment.

6 Grading scale

- Grades are A, B, C, D, F (no $\pm$ grades). The final grade will be curved to ensure fair/homogeneous grading between sections.
- Grades are based on lab reports and every report is weighted equally for the final grade.
- After the grades are normalized across all sections $90 \leq A \leq 100$ and $80 \leq B < 90$ and $70 \leq C < 80$ and $60 \leq D < 70$.

7 AI policy

Any use of Artificial Intelligence or Large Language Models (AI/LLM) to generate parts of the pre-lab or lab report, will be considered a blatant attempt at cheating and will result in a failing grade in the course.